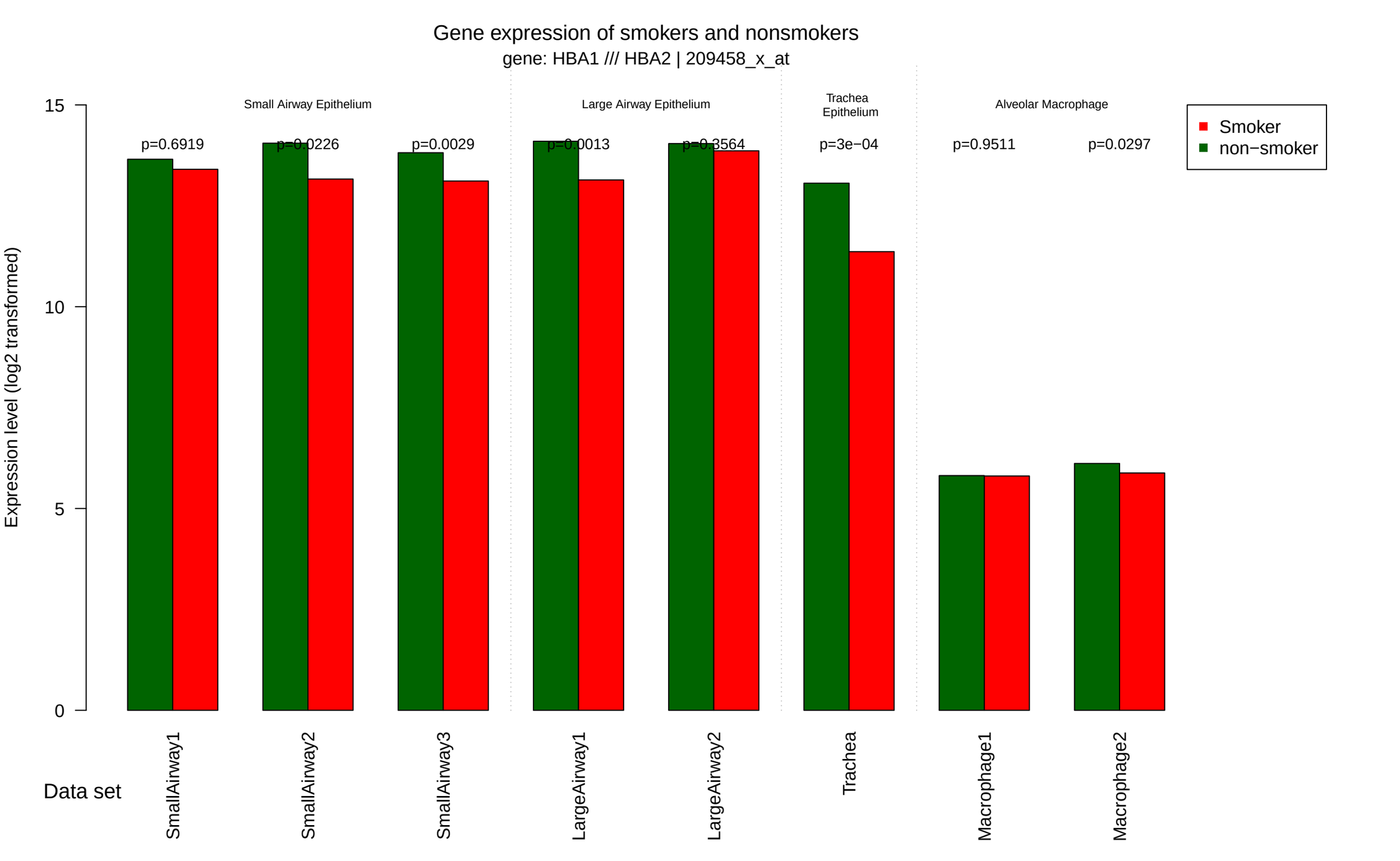
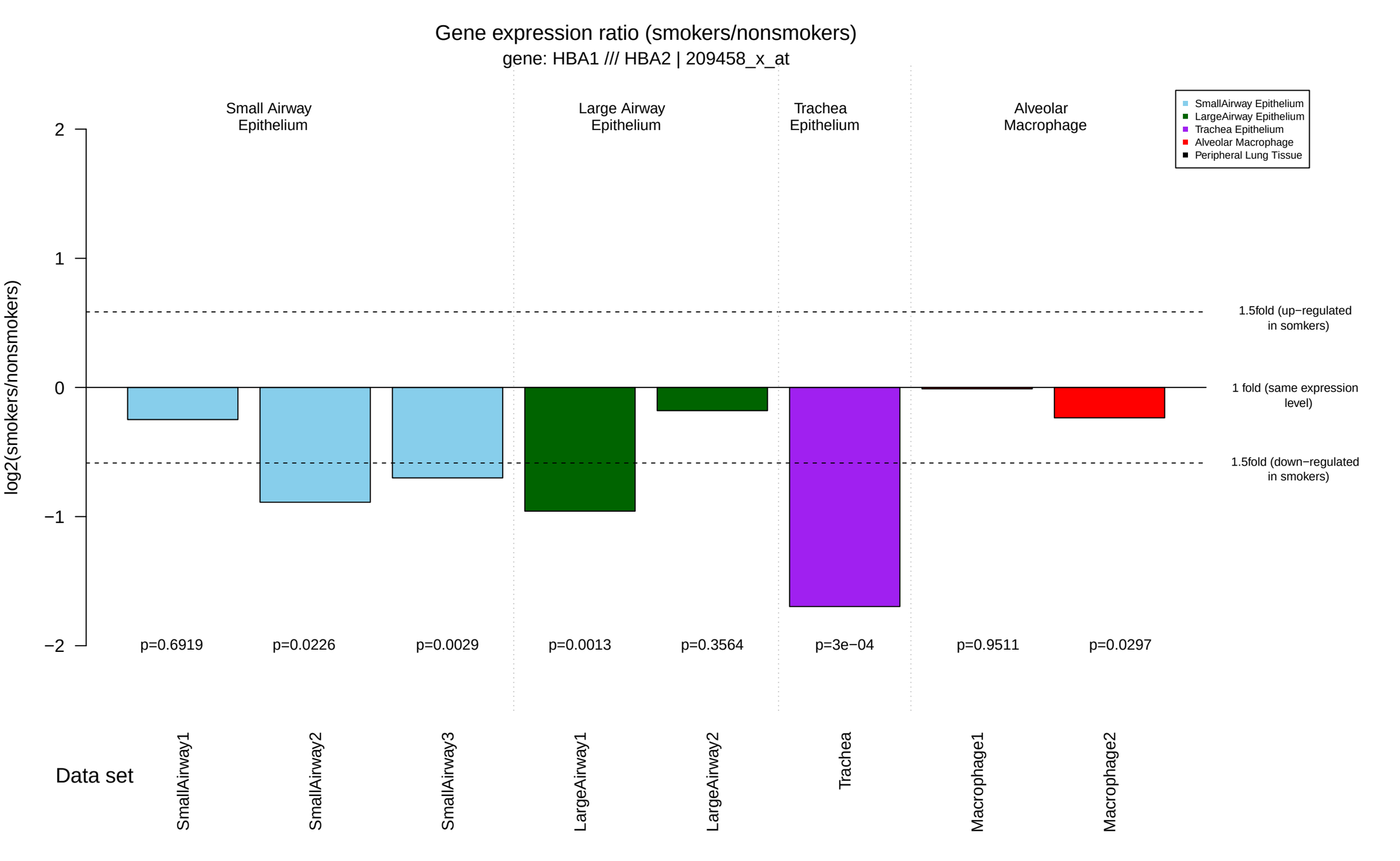
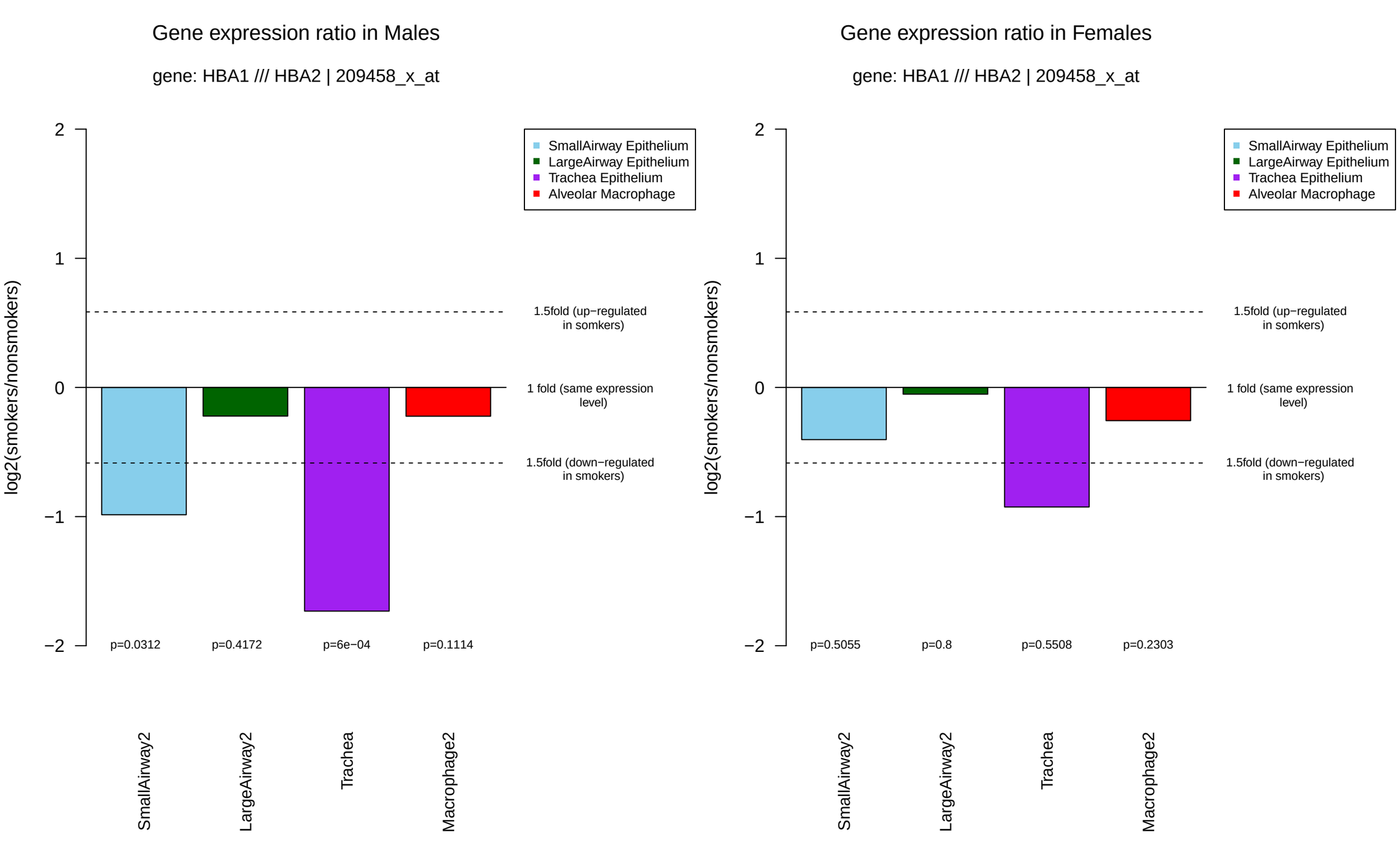
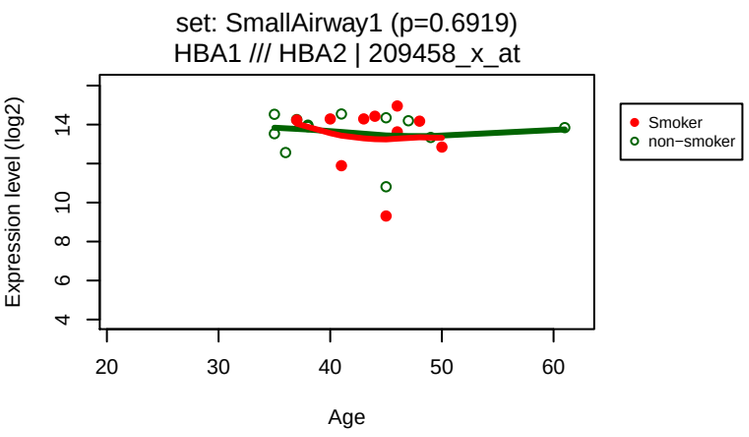
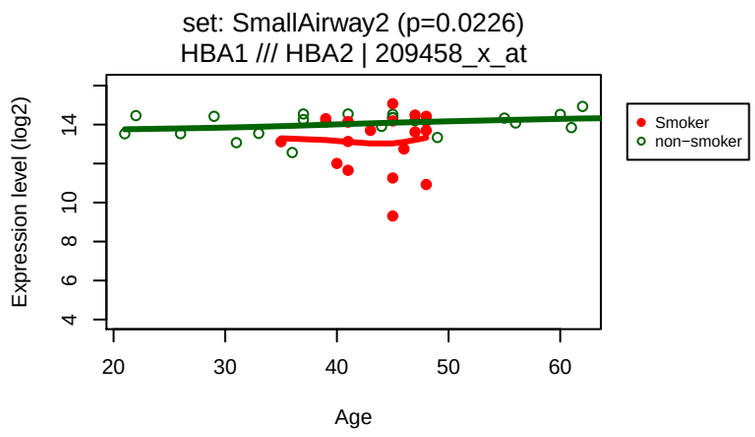
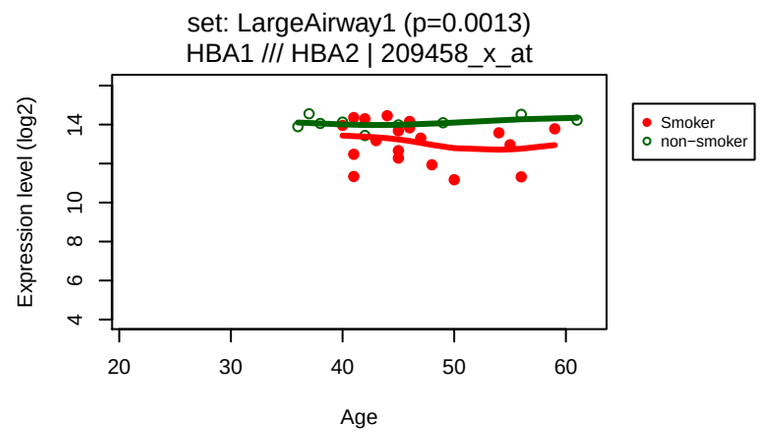

Gene expression of smokers and nonsmokers
gene: HBA1 /// HBA2 | 209458_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6919
p=0.0226
p=0.0029
p=0.0013
p=0.3564
p=3e−04
p=0.9511
p=0.0297
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

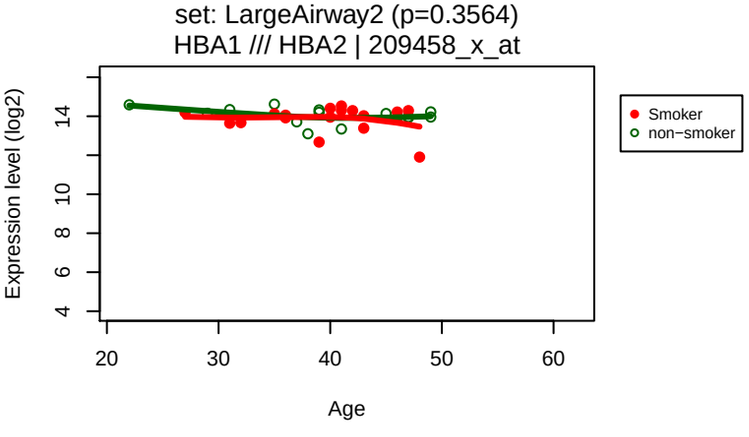

Gene expression ratio (smokers/nonsmokers)
gene: HBA1 /// HBA2 | 209458_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6919
p=0.0226
p=0.0029
p=0.0013
p=0.3564
p=3e−04
p=0.9511
p=0.0297
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

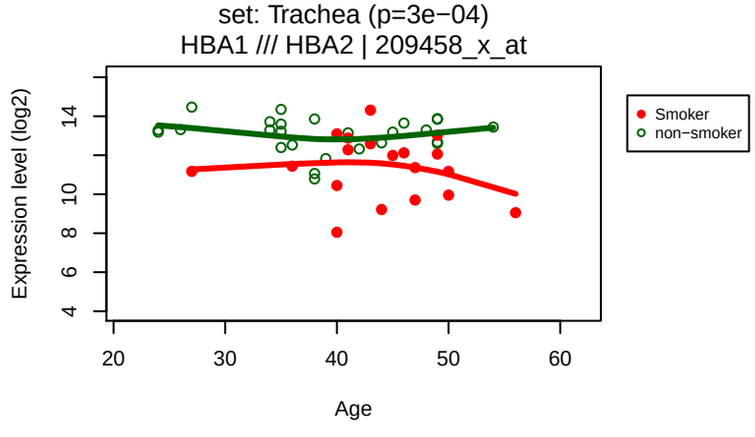

Gene expression ratio in Males
gene: HBA1 /// HBA2 | 209458_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0312
p=0.4172
p=6e−04
p=0.1114
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HBA1 /// HBA2 | 209458_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5055
p=0.8
p=0.5508
p=0.2303
SmallAirway2
LargeAirway2
Trachea
Macrophage2

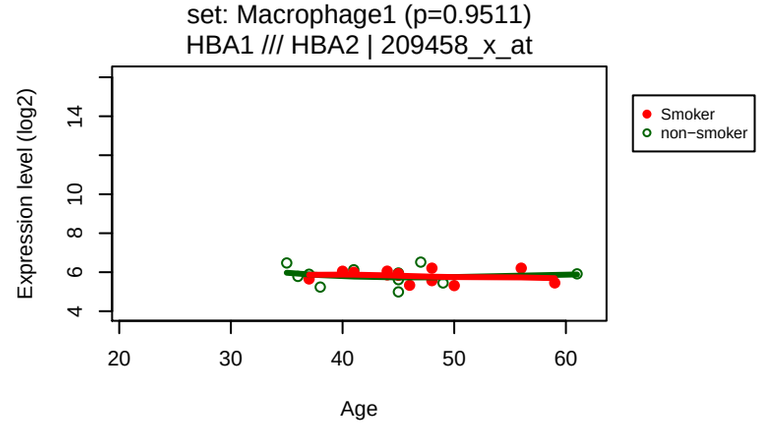

set: SmallAirway1 (p=0.6919)
HBA1 /// HBA2 | 209458_x_at
Expression level (log2)
Age
Smoker
non−smoker

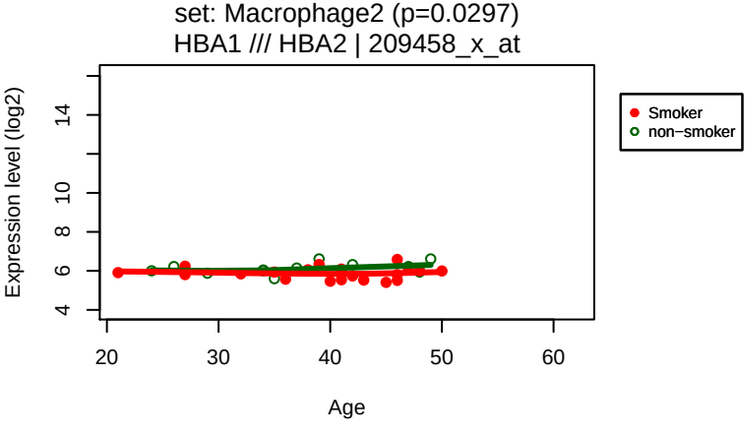

set: SmallAirway2 (p=0.0226)
HBA1 /// HBA2 | 209458_x_at
Expression level (log2)
Age
Smoker
non−smoker

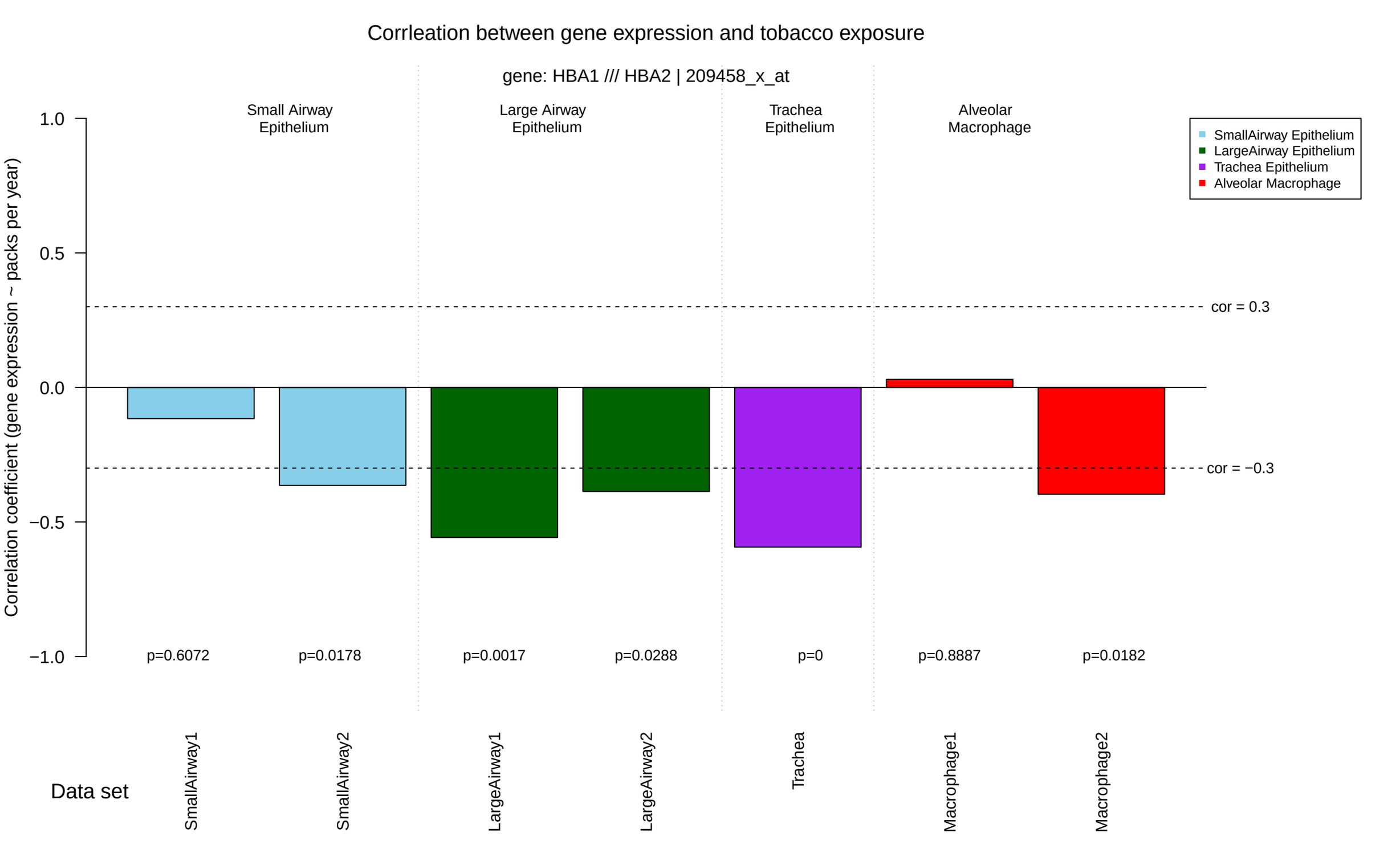

set: LargeAirway1 (p=0.0013)
HBA1 /// HBA2 | 209458_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3564)
HBA1 /// HBA2 | 209458_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=3e−04)
HBA1 /// HBA2 | 209458_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9511)
HBA1 /// HBA2 | 209458_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0297)
HBA1 /// HBA2 | 209458_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HBA1 /// HBA2 | 209458_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6072
p=0.0178
p=0.0017
p=0.0288
p=0
p=0.8887
p=0.0182
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2